Step 10

⇒ Flight Path Angle Calc. for Rocket Launch

- $\gamma = \operatorname{asin}\left(\frac{\vec{v}\cdot\vec{r}}{\|\vec{v}\|\,\|\vec{r}\|}\right) = \frac{\pi}{2} - \operatorname{acos}\left(\frac{\vec{v}\cdot\vec{r}}{\|\vec{v}\|\,\|\vec{r}\|}\right)$

where
$\vec{v}$ is either:
$\vec{v}_{inertial}$, used above 80 km
or
$\vec{v}_{rel} = \vec{v}_{inertial} - (\vec{\omega}_{\oplus} \times \vec{r})$, used below 40 km

- Can apply fxn to blend use of $\vec{v}_{inertial}$ & $\vec{v}_{rel}$ betw 40 km & 80 km:

- To implement continuous blending w/o sudden steering "jerk", use smooth step fxn based on altitude (h):
use Cubic hermite spline to ensure smooth derivatives (velocities & accelerations) at transition edges

$\ast$ $\vec{v}_{steer}(h) = (1 - w(h))\,\vec{v}_{rel} + w(h)\,\vec{v}_{inertial}$
where
$w(h)$ = weight factor in range $[0, 1]$
Define transition window using lower alt. limit (h_{min}) & upper alt. limit (h_{max})
Standard Earth parameters:
$h_{min} = 40$ km
$h_{max} = 80$ km
Normalize alt. into $[0, 1]$ range:
$$x = \frac{x - h_{min}}{h_{max} - h_{min}}$$
Clamp x to ensure its within $[0, 1]$ range:
$x = \max(0, \min(1, x))$

(Formulation of $W(h)$ in $\vec{V}_{steer}$ eqn cont.)

Apply the cubic hermite spline to get $w(h)$:

$$w(h) = 3x^2 - 2x^3$$

This ensures $\frac{dw}{dh} = 0$ at both h_{min} & h_{max}.

> Note: (E.g. q, F_D, M)

Always compute aerodynamic physics (E.g. q, F_D, M) strictly using the relative wind frame.

Blending fxn used to compute $\vec{V}_{steer}$, used in computing γ (& the η required to achieve γ at burnout)